Chapter 11. Variable Selection

Consequences of omitting variables, search methods, selection criteria, Mallows' C_p

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With many candidate regressors we want a **parsimonious** model: enough variables to explain y well and predict accurately, but no more, since every extra variable adds variance to the estimates and predictions. This chapter quantifies the trade-off and describes search methods and criteria.

11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where X_k $(n \times (k+1))$ contains the intercept and k regressors and X_r $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

Theorem 11.1

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

Proof. (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var +bias $\cdot$ bias$'$.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var +bias^2 with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

Theorem 11.2

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

Proof. By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

In words: dropping variables whose coefficients are small relative to their standard errors (so that $\beta_r\beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces the mean squared error of estimation and prediction, even though it introduces bias. Conversely, if the reduced model is true ($\beta_r = \mathbf{0}$), both estimators are unbiased and the reduced one has smaller variance, so including irrelevant variables only costs precision.

11.2 Search methods

With p candidate regressors there are 2^p subsets (each variable in or out), so exhaustive search becomes infeasible for large p ($2^{50} \approx 10^{15}$).

All possible regressions (best subset). Fit all 2^p models. For each size k, keep the best model by R_k^2 (equivalently by SSE_k or MSE_k, since these are monotone in each other for fixed k), then compare the best models across k with a criterion that penalizes size (Section 11.3).

Backward elimination.

(1) Fit the full model.

(2) Compute the partial F statistic of each variable in the model.

(3) Let F_L be the smallest, and F_C the critical value at level α_{drop}.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

Forward selection. Start with no regressors. At each stage add the variable that increases R^2 the most (equivalently, has the largest partial F given those already in), provided its partial F exceeds the critical value at level α_{add}; otherwise stop.

Stepwise selection. Forward selection, but after each addition the variables already in the model are re-examined by partial F tests and removed if they fall below the α_{drop} threshold (with $\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling). A variable that was useful early can become redundant once others enter.

The methods need not agree, and none guarantees the best subset. The levels used are not true significance levels, because of the repeated testing.

Example 11.1

In Hald's classical cement data ($n = 13$, four ingredients $x_1, \ldots, x_4$), x_2 and x_4 are highly correlated ($x_1 + x_2 + x_3 + x_4 \approx 100$). Backward elimination ($\alpha = 0.05$) removes x_3 (partial $F = 0.02$) and then x_4 (partial $F = 1.86 < F_{0.05}(1, 9) = 5.12$), ending with $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters x_4 first (largest single $R^2 = 0.675$), then x_1, and stops because x_2 has partial $F = 5.03 < 5.12$, ending with $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise selection with $\alpha_{\text{add}} = 0.10$ enters x_2 and then removes x_4 (partial $F = 1.86$), ending with $\{x_1, x_2\}$. Because x_2 and x_4 carry nearly the same information, different routes end at different, almost equally

good models.

11.3 Selection criteria

Motivation. Adding variables never reduces R^2, so it cannot choose a model size. Criteria such as adjusted R^2, Mallows' C_p, AIC and PRESS penalize complexity or estimate prediction error directly.

For a model with k regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R^2_k = 1 - SSE_k/SST$;

(3) adjusted R^2: $R^2_{a,k} = 1 - \dfrac{n-1}{n-k-1}(1-R^2_k) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the MSE of the full model.

R^2_k never decreases when a variable is added, so it can only compare models of the same size. MSE_k can increase when a variable is added (the loss of one degree of freedom may outweigh the decrease in SSE), and $R^2_{a,k}$ is a decreasing function of MSE_k: maximizing R^2_a is minimizing MSE_k.

Mallows' C_p. We judge the reduced model by the total standardized mean squared error of its fitted values at the n design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}'_{ki}\tilde\beta_k.$$

Theorem 11.3

$$\Gamma_k = (k+1) + \frac{\beta'_r X'_r (I-H_k) X_r \beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence C_k estimates Γ_k, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

Proof [Supplement]. By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}'_{ki}(X'_kX_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X'_kX_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}'_{ki}\Big\} = \mathrm{tr}\{(X'_kX_k)^{-1}X'_kX_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta'_rX'_r(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ■

So we look for models with small C_k and C_k close to $k+1$; a C_k well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.

11.4 Selected exercises

Motivation. The first exercise makes the omitted-variable bias explicit for a straight line; the others link the selection criteria to the partial F statistic and derive AIC for the normal linear model.

Exercise 11.1 (Omitted-variable bias for a straight line)

The straight line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$ is fitted by least squares, but the true mean is $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ (errors uncorrelated with variance σ^2). Write $S_{11} = \sum(x_{i1} - \bar{x}_1)^2 > 0$, $S_{22} = \sum(x_{i2} - \bar{x}_2)^2$ and $S_{12} = \sum(x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$.
(a) Is $\hat{\beta}_1$ biased? Find $E\hat{\beta}_1$.
(b) Find the expectation of the residual mean square of the fitted line.

Solution.

(a) With $k_i = (x_{i1} - \bar{x}_1)/S_{11}$, $\hat{\beta}_1 = \sum k_i y_i$, $\sum k_i = 0$ and $\sum k_i x_{i1} = 1$ (Section 4.1). Hence

$$E\hat{\beta}_1 = \sum k_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) = \beta_1 + \beta_2 \sum k_i x_{i2} = \beta_1 + \beta_2 \frac{S_{12}}{S_{11}}.$$

So $\hat{\beta}_1$ is biased unless $\beta_2 = 0$ or $S_{12} = 0$ (the omitted regressor is uncorrelated with x_1 in the sample). The factor S_{12}/S_{11} is the slope of the regression of x_2 on x_1, the second entry of the alias matrix A of Theorem 11.1: $\hat{\beta}_1$ picks up the part of the effect of x_2 that travels with x_1.

(b) By Theorem 11.1(b) with $k = 1$ and $X_r = \mathbf{x}_2$, the excess is $\beta_2^2 \, \mathbf{x}_2'(I - H_k)\mathbf{x}_2/(n-2)$, where $\mathbf{x}_2'(I - H_k)\mathbf{x}_2$ is the residual sum of squares of the regression of x_2 on $(1, x_1)$, namely $S_{22} - S_{12}^2/S_{11}$ (Theorem 3.3). So

$$E(MSE) = \sigma^2 + \frac{\beta_2^2(S_{22} - S_{12}^2/S_{11})}{n-2} \geq \sigma^2.$$

■

Exercise 11.2 (Adjusted R^2 and the partial F statistic)

A model with k regressors (and intercept) is enlarged by r further regressors, and F is the partial F statistic for the added regressors, computed with the residual mean square of the larger model. Assuming $SSE_{k+r} > 0$, show that the adjusted R^2 increases if and only if $F > 1$.

Solution [Supplement].

Step 1. Since $R_a^2 = 1 - MSE/\{SST/(n-1)\}$ with SST common to both models, R_a^2 increases iff $MSE_{k+r} < MSE_k$.

Step 2. By definition $F = \{(SSE_k - SSE_{k+r})/r\}/MSE_{k+r}$, so $SSE_k = SSE_{k+r} + rF\,MSE_{k+r} = (n - k - r - 1 + rF)MSE_{k+r}$, and

$$MSE_k = \frac{SSE_k}{n-k-1} = \frac{n-k-r-1+rF}{n-k-1} MSE_{k+r}.$$

Step 3. With $MSE_{k+r} > 0$, $MSE_k > MSE_{k+r}$ iff $n - k - r - 1 + rF > n - k - 1$ iff $F > 1$. So adjusted R^2 admits a variable far more easily than a significance test, which would require $F > F_\alpha$ (about 4 for one variable at $\alpha = 0.05$). ■

Exercise 11.3 (Mallows' C_k and the partial F statistic)

Let F_0 be the partial F statistic for dropping the $r = p - k$ regressors of X_r from the full model. Show that

$$C_k = (k+1) + (p-k)(F_0 - 1),$$

so that $C_k \leq k + 1$ if and only if $F_0 \leq 1$.

Solution [Supplement]. Let $\hat{\sigma}^2 = SSE/(n-p-1)$ be the full-model residual mean square. By definition of F_0, $SSE_k = SSE + (p-k)F_0\hat{\sigma}^2 = \{(n-p-1) + (p-k)F_0\}\hat{\sigma}^2$. Substituting into $C_k = SSE_k/\hat{\sigma}^2 + 2(k+1) - n$,

$$C_k = (n-p-1) + (p-k)F_0 + 2(k+1) - n = (k+1) + (p-k)F_0 - (p-k).$$

Since $F_0 \approx 1$ on average when $\beta_r = \mathbf{0}$ (its mean is $\nu/(\nu-2)$ with $\nu = n-p-1$), this matches $C_k \approx k+1$ in Theorem 11.3. ■

Exercise 11.4 (AIC for the normal linear model)

For a model with k regressors and intercept and $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, define $\mathrm{AIC}_k = -2\,l_{\max} + 2(\text{number of parameters})$, where $l_{\max}$ is the maximized log-likelihood and the parameters are β_k and σ^2. Show that

$$\mathrm{AIC}_k = n \log \frac{SSE_k}{n} + 2(k+1) + n(\log 2\pi + 1) + 2,$$

so that, up to a constant common to all models, $\mathrm{AIC}_k = n\log(SSE_k/n) + 2(k+1)$.

Solution [Supplement].

Step 1. The log-likelihood is $l(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X_k\beta\|^2$. For each σ^2 it is maximized over β by the least squares estimator (Theorem 6.1), giving $-\frac{n}{2}\log(2\pi\sigma^2) - SSE_k/(2\sigma^2)$.

Step 2. Assume $SSE_k > 0$ (true with probability one). The derivative in σ^2 is $-\frac{n}{2\sigma^2} + \frac{SSE_k}{2\sigma^4}$, positive for $\sigma^2 < SSE_k/n$ and negative beyond, so the maximum is at $\hat{\sigma}^2 = SSE_k/n$, and

$$l_{\max} = -\frac{n}{2}\log\frac{2\pi SSE_k}{n} - \frac{n}{2}.$$

Step 3. There are $k+2$ parameters, so $\mathrm{AIC}_k = n\log(SSE_k/n) + n\log 2\pi + n + 2(k+2)$, which is the claimed expression. Like C_k, it trades the decrease of SSE_k against a penalty of 2 per parameter. ■